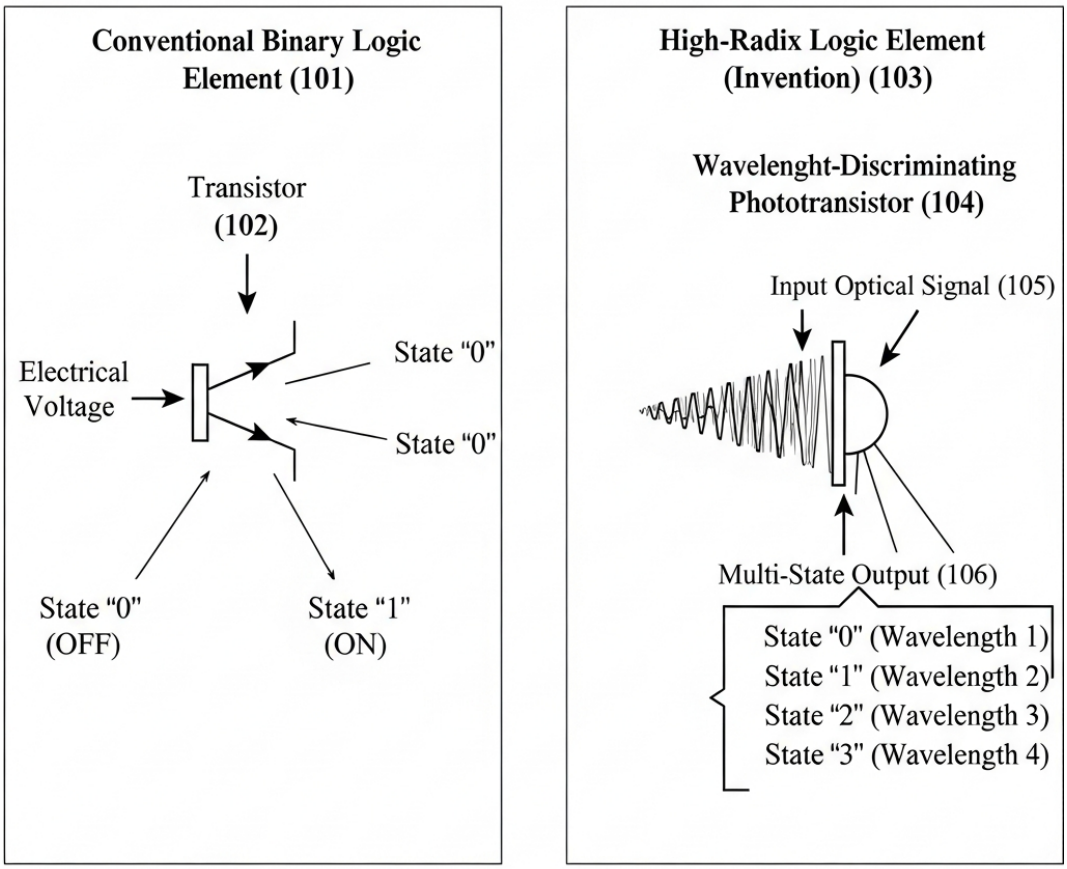

Conventional Binary Logic Element (101)
Transistor (102)
Electrical Voltage
State "0"
State "0"
State "0" (OFF)
State "1" (ON)
High-Radix Logic Element (Invention) (103)
Wavelenght-Discriminating Phototransistor (104)
Input Optical Signal (105)
Multi-State Output (106)
State "0" (Wavelength 1)
State "1" (Wavelength 2)
State "2" (Wavelength 3)
State "3" (Wavelength 4)

FIG. 1

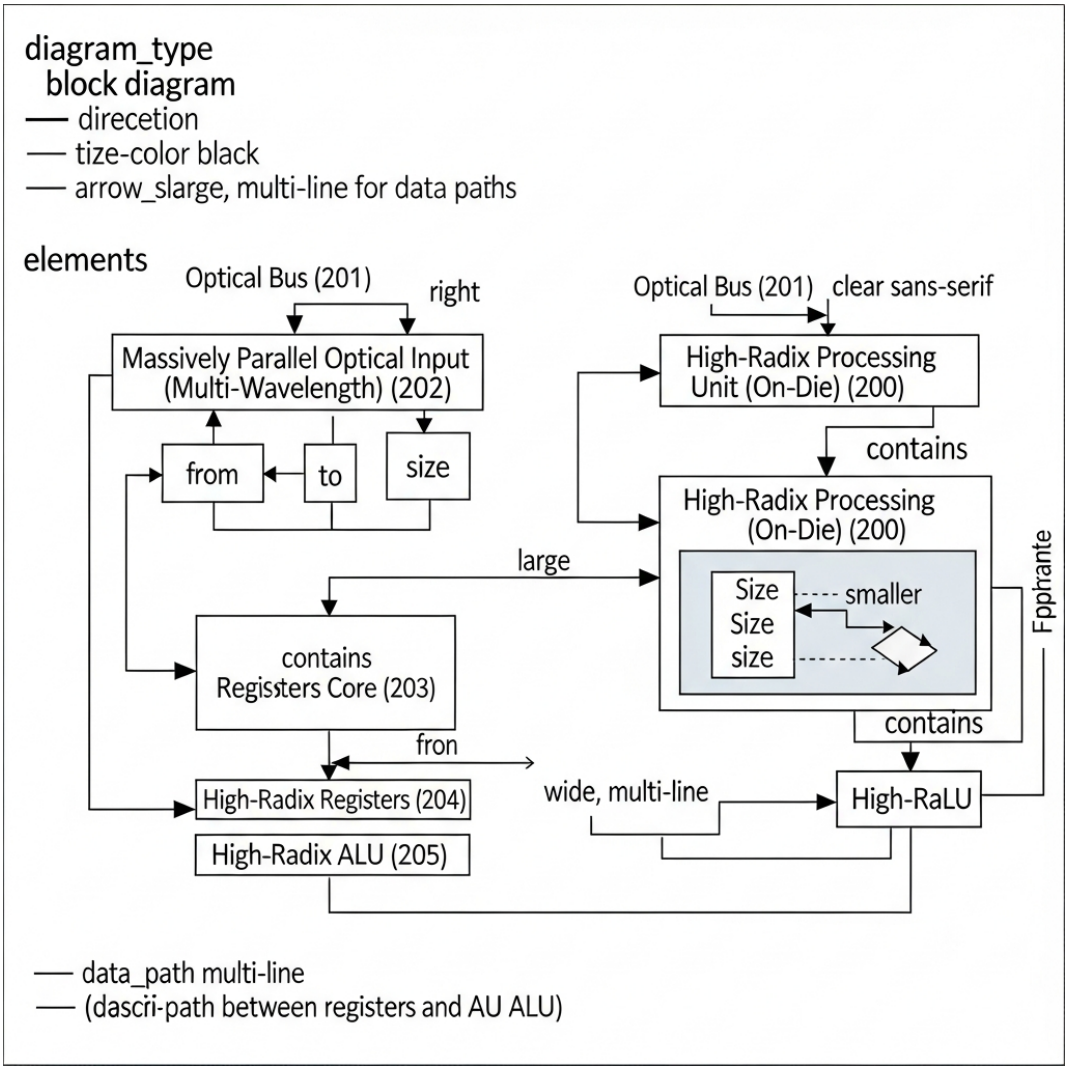

diagram_type
block diagram
— direcetion
— tize-color black
— arrow_slarge, multi-line for data paths
elements
Optical Bus (201)
right
Massively Parallel Optical Input (Multi-Wavelength) (202)
from
to
size
Optical Bus (201)
clear sans-serif
High-Radix Processing Unit (On-Die) (200)
contains
High-Radix Processing (On-Die) (200)
large
Size
Size
size
smaller
Fpprante
contains Regisssters Core (203)
contains
fron
High-Radix Registers (204)
High-Radix ALU (205)
wide, multi-line
High-RaLU
— data_path multi-line
— (dascri-path between registers and AU ALU)

FIG. 2